
SOIL PHYSICS

Soil Properties as Indicators of Soil Water Regime: A Review

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Abstract—Water movement in soils contributes to the formation of a number of specific soil properties; their interpretation makes it possible to characterize the features of particular hydrological processes and the soil water regime in general, which is important because of the limited possibilities to conduct direct monitoring of soil water movement. In this review, we consider the diagnostic indicators of mineral soils determined in the field and under laboratory conditions and characterized by different formation times and tolerance toward changes in the environmental factors. Field methods for diagnosing soil water regime make it possible to formulate hypotheses about the features of water behavior within the soil profile, which can later be confirmed or refuted as a result of laboratory diagnostic methods, regime observations, and physical and mathematical modeling. In particular, on the basis of a standard field description of soils, it is possible to identify the zone of maximum water turnover, parts of the soil profile with constant or periodic water stagnation, and the capillary fringe level. Clay coatings, root channels, and burrows of soil animals mark the preferential flows. Laboratory diagnostic methods are aimed at assessing the quantitative ratio of hydrological processes in the soil; basically, they allow to characterize the source, duration, and intensity of the period of soil overmoistening. The Schwertmann coefficient is the most commonly used and efficient analytical indicator for diagnosing soil overmoistening, which has been shown for a wide range of soils. Verification of the results of hydrological modeling on the basis of data on soil properties is potentially possible for the processes of downward and lateral runoff, physical evaporation, transpiration, evapotranspiration, and capillary rise of soil water.

Keywords: hydropedology, soil overmoistening, hydrological simulation, water regime, gley

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INTRODUCTION

Movement and redistribution of water in landscape and soil are the most important mechanisms of the radial and lateral differentiation of soils and determine the diversity of edaphic and agroecological conditions of the territory [44, 67, 87]. However, the study of the direction and intensity of hydrological processes in soils is associated with a number of problems; the main ones are the limited possibilities for organizing long-term instrumental observations of the seasonal and long-term variability in the water regime of soils [57], as well as the processes of lateral soil water migration in the landscape [68, 116]. Despite advances in the development of direct methods for monitoring the soil water regime [61, 111] and lateral water movement [75, 84, 90], comprehensive landscape-hydrological studies are few [74, 90].

An alternative approach to the study of hydrological processes in soils is associated with their simulation [57, 85, 90]. In the recent decades, some progress has been made in this direction; a review of landscape hydrological models, their input parameters, and the

analysis of modeling results can be found in recent works [74, 105, 115].

A special problem of modeling the behavior of water in soils is the verification of the obtained results; its solution implies comparing the modeled values with (1) the results of regime observations of soil moisture, or with (2) indirect indicators of the soil and/or landscape moisture distribution [74, 94, 105, 114]. These indicators include both the soil formation products themselves [64, 72, 89] and natural and anthropogenic tracers (for example, pesticides, herbicides, water isotopes, etc.). The latter are used in the absolute majority of works on verification of the results of soil-hydrological modeling [62, 109, 112]. At the same time, the use of tracers does not allow verifying the results of modeling water movement in soils over a period of more than a century [111]. The advantage of using the profile (and/or landscape) distribution of soil-forming products as indicators of water movement in the soil is not only a wide spatial and temporal range (from several seconds to millennia, depending on the indicator used) but also the accompanying interpretation of the genesis of soils and their particular properties [96].

The aim of this work is to review, analyze, and genetically interpret the diagnostic indicators of hydrological processes in mineral soils, including those that are used in parameterization and verification of simulation models of the water regime for different time intervals. The main hydrological processes in soils include: moisture absorption, downward and lateral flows, capillary rise, thermal moisture transfer, water transpiration by plants, physical evaporation, freezing, thawing, vapor transfer, and moisture condensation. All of the listed processes determines the water regime of soils, and the quantitative characteristic of the volumes of water involved in these processes determines the soil water budget [58]. Under the parameters of the water regime, we understand any of its properties, for example, the duration of waterlogging, the filtration rate, the level of fluctuation of the capillary fringe, the areas of descending and ascending water flows, etc. Diagnostic indicators can characterize both the water regime of soils as a whole and individual hydrological processes in soils. These indicators can be used both to verify the vertical (within the soil profile) and lateral (in the landscape) movement of moisture and are based on the same principle—comparison of the calculated values with their quantitative or qualitative indicators within the soil profile or associated soil areas.

The structure of the review includes a consistent consideration of diagnostic indicators determined in the field (soil color and color patterns; redox potential and activity of ions in soil solutions; morphological characteristics of pedofeatures; classification position of soils in the systems of *Classification and Diagnostics of Soils of the USSR* [34] or in *Soil Taxonomy* [82]) and laboratory conditions (indicators of soil moisture and air porosity; forms of occurrence and content of iron in soils; chemical composition of nodules; mineralogical composition of soils; characterization of soil organic matter; isotopic composition of soil water). At the end of the review, we give a brief comparative description of the indicators.

FIELD METHODS FOR DIAGNOSING SOIL WATER REGIME PARAMETERS

Color patterns. Cold shades (gray, greenish, gray-blue) are traditionally interpreted as the result of soil waterlogging and the presence of Fe^{2+} compounds; in some cases, as the features inherited from soil-forming materials rich in minerals specific in color (for example, some phyllosilicates). A qualitative reaction to the presence of Fe^{2+} is the appearance of a reddish color after spraying a freshly cut soil section with a solution of 0.2% α, α -dipyridyl (may not be applicable in neutral and alkaline soils), or a dark blue color in the presence of potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$) [77].

Munsell color chart is most commonly used to quantify soil color. According to WRB [77], the color

of the gley horizon corresponds to the colors of sheets Gley 1 and Gley 2, N, 10Y, GY, G, BG, B, and PB; or sheets 2.5Y and 5Y at chroma values ≤ 2 in the case of a uniform color of the horizon; or a hue 2.5 units redder and a chroma ≥ 1 unit higher (in moist state) than those in the surrounding material or in the underlying layer. In the work of Targulian with coauthors devoted to gley morphotypes [50], it was shown that a uniform gley coloration of the horizon in the absence of iron and manganese concentrations and nodules is indicative of the prolonged waterlogging of soils with filling of soil pores with water. In the case of a heterogeneous color pattern—the presence of brown, red, and ochreous zones and segregational pedofeatures (but not solid ones)—we can talk about a combination of periods of waterlogging and desiccation gradually replacing one another. The presence of solid segregational forms indicates a sharp change in the moisture conditions of the horizon. A detailed description and classification of gley horizons by color patterns and coloration is given in the work of Zaidel'man [19]; in total, three gleyic horizons and five gley horizons were identified. According to Zaidel'man [19], marbled (mottled), banded, and localized gley patterns are always the result of surface (not ground) waterlogging. In addition, the work gives a hydrological interpretation of the formation of various light-colored acidic (eluvial) podzolic horizons (albic material) [19].

Color determination according to Munsell color chart can be used to interpret the mineralogical composition of nodules formed under different moistening conditions [101]. The pretreatment of samples with solutions of hydrogen peroxide and oxalates is recommended to remove color-masking compounds (organic carbon, manganese). Thus, orange colors with a hue of 7.5YR and a value of more than 6 can be due to the presence of lepidocrocite, a mineral that forms when anaerobic conditions are replaced by aerobic conditions in carbonate-free soils. Ferrihydrite (5–7.5YR, value < 6) is formed during a sharp desiccation of waterlogged horizons.

Evans and Franzmeier [70] developed two soil color indices based on the Munsell soil color book to identify waterlogged Indiana soils and make decisions about the need for soil drainage. Index C1 is based on the use of chroma; index C2, on hue and chroma. When calculating the indices, the color parameters of the main mass of each horizon, as well as the colors of mottles and coatings (if any) are taken into account; the values of C1 and C2 indices are weighted average (relative to the thickness of the horizons) values of corresponding indices in soil horizons. The lower the index value, the more waterlogged the soil. Index C1 was successfully used to characterize the duration of soil waterlogging at depths of 50–100 and 100–150 cm in Taiwan [78].

In addition to the Munsell soil color book, the CIE-L*a*b* system [14] can be used to characterize

soil color numerically; color determination in this system is possible in the field using a smartphone (App Color Muse) or portable devices such as Nix Pro cameras [110]. Vodyanitskii and Shishov [15] proposed a method for diagnosing waterlogged soils by the increase in a^* values (redness–greenness) after the soil treatment with an H_2O_2 solution. Automorphic soils are characterized by an increase in a^* after the treatment, while hydromorphic soils are characterized by the opposite tendency. Another coefficient developed using this color system is the calculation of the conditionally red pigment (Hem cond.) according to the formula [13]:

$$\text{Hem cond.} = \frac{\frac{\sqrt{a(a^2 + b^2)}}{7b} - 0.54}{1.97},$$

where a and b are the values along the corresponding axes in the CIE- $L^*a^*b^*$ color system. Using the example of soils in the Cis-Urals, it was shown [13] that Hem cond in waterlogged horizons is less than 0.1.

Thus, the advantage of using soil color characteristics to verify the results of hydrological modeling is the ease of determining indicators and fixing information about soil color in a standard soil description, which opens up the possibility of creating databases on materials available for almost any territory. Based on the color, it is possible to identify areas with constant and periodic waterlogging and to rank the studied soil layers (soil areas) according to the duration of waterlogging (from less to longer waterlogged areas).

At the end of this part, we note that the result of determining the color of soils depends on the lighting conditions, the water content of the sample, the characteristics of the color perception of the researcher (when visually determining the color according to the Munsell soil color book), or the device used in the instrumental approach, which affects the accuracy of the result. In [92], an analysis of the results of the reproducibility of determining the color of soils visually, using inexpensive portable devices (Cube, Nix Pro, Color Muse) and professional colorimeters was given. It was shown that the most economic and efficient way is to use a Nix Pro camera, which allows one to determine color parameters in various systems, including the Munsell and the CIE- $L^*a^*b^*$ systems.

Partial pressure of hydrogen and activity of ions in soil solutions. The filling of pores with water leads to a change in the redox conditions of soils; therefore, the parameters of the redox state of soils can act as indicators of hydrological processes in soils. These indicators include the values of the redox potential (Eh), which reflects the activity of ions in soil solutions, and the partial pressure of hydrogen (rH) [9]. The determination of the redox (oxidation–reduction) potential of soils in the field is carried out with special portable devices. It has been shown that at values of the redox potential above 450 mV, the soils are characterized by

the oxidizing medium and the absence of waterlogging; at values below 200 mV, a pronounced reduction is observed, and the soils are waterlogged [17].

According to WRB [77], the diagnostics of reducing conditions in waterlogged soils and horizons is based on the calculation of the values of the partial pressure of hydrogen (rH):

$$rH = Eh \times 29^{-1} + 2 \times pH$$

at $rH < 20$, reducing conditions are diagnosed in the soil.

The properties of the liquid and gas phases of soils quickly come into balance with the characteristics of the environment. Therefore, the diagnostic indicators of the redox regime of soils characterize the result of hydrological processes in a short time period [9]. The rapid response of Eh and rH values to changes in the environmental conditions can be expressed in the fact that soils experiencing strong spring waterlogging may not differ in Eh and rH values from soils that do not experience spring waterlogging, if the measurements are carried out in a dry summer period. For example, Zaidel'man [27] demonstrated that gley soils forming under conditions of deep groundwater table can be characterized by high values of the redox potential in summer, often close to its values in nongleyed automorphic soils.

Note that the considered indicators can be a reflection of changes not only in soil moisture but also in the formation of soil sulfides (sulfidogenesis). Therefore, the use of Eh and rH values for the purpose of diagnosing waterlogging in soils with a high sulfur content or in soils of the zone of influence of sulfate waters and hydrogen sulfide emanation should be carried out with caution [56].

Pedofeatures. Pedofeatures of metals with variable valence. Coatings. Iron, iron–manganese, and manganese nodules and concretions are formed during a combination of processes of pronounced waterlogging and desiccation; thus, the area of the soil profile, which is characterized by the presence of these pedofeatures, should experience a stagnant–percolative soil water regime (either modern or relict). The factor that determines the number of nodules is the duration of the anaerobic period. During short-term moisture stagnation, concretions undifferentiated in color are formed, and during long-term moisture stagnation, a Mn-containing core and a light brown ferruginous outer part are clearly distinguished in nodules [25].

The formation of pseudofibers (ortsteins)—dense, thin subhorizontal interlayers cemented with iron compounds in coarse-textured soils—is associated with changes in the level of groundwater and features of lithogenesis or illuviation of solutions saturated with iron oxides under present-day or relict conditions [20, 48, 99]. Rudyak (nodulous, iron-ore, pisoplinthic horizons) in the form of a solid slab or a conglomerate of nodules occupying >50% of the area of the vertical

Table 1. Carbonate pedofeatures and hydrological conditions of their formation [54]

Pedofeature	Variety	Conditions of formation
Efflorescence	Separate fine carbonate threads closely associated with the soil mass	Upon drying of the soil surface, from solutions of low concentration
Coatings	Fibrous—cotton-like mass of tangled fibers in voids	Low concentration of solutions and their quick evaporation
	Dense—thin homogeneous layer of CaCO_3 on ped faces	More concentrated solutions and gradual drying of the soil
	Crusts—thicker (up to 1–5 cm) and dense coatings of the walls of macrocracks	No data
Veins	Weakly cemented relatively large grains of carbonates	In spring, with downward migration of soil solutions with a low concentration of carbonates
	Incrustation along pore walls with a dense homogeneous mass of CaCO_3	At the beginning of the growing season, from concentrated solutions that completely fill the pores
	Columns—voids fully filled with CaCO_3	Movement of carbonates with film water
	Veins surrounded by a halo of carbonate impregnation of the soil mass	Infiltration of solutions from the soil mass into the pores and precipitation on pore walls and in the adjacent soil mass
Soft nodules (white eyes)	Incretions—yellowish, with indistinct shape and a significant admixture of clay material	Gradual concentration of carbonates in the soil mass around pores and voids
	Accretions—clearly separated from the host mass and containing a minimum amount of clastic grains	Evaporation of film water from its accumulations at the joints of peds into large cracks
Loess dolls		Precipitation of calcite during the movement of colloidal solutions through capillaries

cut are formed as a result of changes in the depth of the groundwater level, or when lateral soil water flows saturated with ferrous forms of iron are discharged onto the surface with a sharp change in the redox regime [11, 21]. Pseudofibers and rudyak horizons act as a barrier to moisture migration, so in the spring they can lead to the formation of perched water; under pseudofibers, a zone of low water content is often diagnosed [20].

Soil coatings—pedofeatures forming on the surface of aggregates and differing from their inner part in morphology and material composition—significantly prevent the penetration of moisture into peds [7]. Coatings, like root and animal burrows, mark the zones of preferential flows in soils [90, 104].

Thus, the use of pedofeatures with polyvalent metals changing their state in dependence on the redox conditions of the medium as indicators of the water regime can make it possible to identify areas with a contrasting change in moistening conditions (the presence of periods of obvious waterlogging and desiccation). Comparison of the number and morphological features of nodules in the studied soils allows us to rank the objects according to the duration of the waterlogging period.

Carbonate pedofeatures. Pedogenic carbonates are widely distributed in soils of semihumid, semiarid, and arid regions. Their size, shape, depth of detection, and the nature of their distribution along the profile are indicators of the soil moisture regime—modern and past [38, 119].

Khokhlova [54] summarized a vast amount of her own and published materials on carbonate pedofeatures and their genesis (Table 1). Interpretation of the morphology of pedofeatures allows us to make assumptions about the direction of water movement in the soil leading to their formation and the rate of evaporation of soil solutions.

In works [28, 30], on the example of forest-steppe and steppe soils, it was revealed that the depth of formation of large carbonate nodules (loess dolls) corresponds to the upper limit of the maximum rise of hard bicarbonate-calcium groundwater. It was shown that these large pedofeatures could form in automorphic soils as a result of climatic changes. A comparison of field descriptions of the same automorphic soils studied in the area of the mown steppe in Kursk region and in the agrolandscape of Belgorod region in 1950–1960 and in the modern period demonstrated that over a period of 40–50 years, signs of hydromorphism in the form of olive-colored horizons and large carbonate nodules (loess dolls) appeared in the lower part of the

profiles of these soils [5]. The authors attribute this phenomenon to the periodic moisture stagnation in soils (the formation of perched water) caused by climate change—an increase in the amount of precipitation and rise of temperatures in the cold season since the early 1970s.

Not only the type of carbonate pedofeatures but also the depth of soil effervescence upon reaction with 10% HCl can be an indicator of the features of the water regime of soils. Thus, differences in the position of the effervescence line correlated with the parameters of the hydrological processes simulated by the HYDRUS-1D model for two agrochernozems under corn and under fallow for a long time [3]. As shown in [89], the depth of soil effervescence correlates with the calculated layer of precipitation redistribution obtained by SIMWE model. According to [5], an increase in the amount of precipitation can lead to both an increase and a decrease in the depth of the effervescence, which is determined by the thickness of the zone of active moisture turnover and the position of the effervescence line in the previous dry period.

In [5], the types of the carbonate profile of chernozems corresponding to different hydrothermal regimes of soil functioning were identified on the basis of the proportion between two zones with carbonates in the soil profile: (a) the zone with labile migrational carbonates effervescing with 10% HCl, but having only finely disseminated carbonates, or patchy carbonate coatings and carbonate threads and (b) the accumulative zone with more stable carbonate concentrations, such as carbonate pseudomycelium and carbonate nodules. The migrational zone corresponds to the zone of maximum moisture turnover, it is characterized by both descending and ascending water flows. In total, three types of spatial relationships between these zones were identified:

(1) The first type is characterized by the presence of a well-defined migrational zone (with a dynamic effervescence line) and an accumulative zone represented exclusively by mycelial (pseudomycelium) forms of carbonate pedofeatures; these zones do not have clear boundaries in the soil profile. This type of the soil profile is characterized by a considerable (about 1 m) depth of spring water recharge, a prolonged presence of mobile moisture below the root layer, a long period of ascending water migration through capillaries, and gradual evaporation of solutions into pores—channels;

(2) In soils of the second type, there is a clearly expressed accumulative horizon with segregational forms of carbonate pedofeatures; the effervescence line is stable in time, and the migrational zone is small or absent. Such soils are characterized by a relatively shallow depth (about 0.5 m) of spring water recharge and rapid drying, which determines the absence of a reserve of mobile moisture in the middle part of the profile and the formation of segregational carbonate

pedofeatures as a result of rapid evaporation of saturated solutions. The upward movement of water through capillaries occurs only within the annual wetting layer and for a short time;

(3) The presence of a deep migrational zone with abundant carbonate “mold” (threads) confined to the middle and lower parts of the humus horizon and underlain by a clearly pronounced accumulative zone (with a predominance of carbonate nodules). Such soils are characterized by deep wetting (down to 2–3 m), the presence of a reserve of available moisture in the middle part of the soil profile, the rise of moisture through capillaries in the spring–summer period, and a sharp soil drying in the second half of summer, which causes the predominance of segregational forms of carbonates in the middle part of the profile).

Note that the formation of segregational forms of carbonate pedofeatures in the second type of the carbonate profile can result not only from the rapid evaporation of saturated solutions but also from a relatively slow evaporation compensated for by the inflow of solutions saturated with calcium carbonate.

The “meadow marl” horizon (calcic horizon) is formed under the conditions of discharge of water flows saturated with carbonates, including lateral flows [27]. This horizon is characterized by a high content of CaCO_3 (>30%) with continuous whitish carbonate impregnation of the soil matrix (carbonate-plugged horizon).

Thus, the interpretation of the features of the carbonate profile (the depth of soil effervescence upon reaction with 10% HCl, the depth of detection and forms of carbonate pedofeatures) makes it possible to characterize the character of soil water recharge, seasonal migration of solutions, the rate and degree of seasonal soil drying, and seasonal contrasts in soil temperature and water content, which can be used in the parameterization and verification of the results of hydrological modeling.

Pedofeatures of gypsum and soluble salts. Pedofeatures of gypsum and soluble salts are found in soils of the steppe, semidesert and desert landscapes, both under automorphic and hydromorphic conditions. Yamnova [59] showed that the form of salt and gypsum pedofeatures is associated not with bioclimatic zoning, but with the composition of groundwater and parent material and characteristics of the modern and relict soil water regimes.

As a rule, the process of gypsum formation in soils is associated with the inflow of groundwater (pore solutions) saturated with Ca and SO_4 . At the same time, gypsum under automorphic conditions can also form due to exchange reactions between the calcium of the soil adsorption complex and sodium sulfate waters, as well as due to decarbonization. In works [59, 60], using the example of the Dhizak station, three morphotypes of the gypsum horizon differing in the conditions of gypsum accumulation and in soil water regime. These

morphotypes are stable over time and remain in the soil profile after a change in the water regime that determines the conditions for their formation:

(1) Incrustation (fine-, medium-, and coarse-crystalline) morphotype in the form of separate and intergrown gypsum crystals. The formation of crystals is confined to the zone of permanent capillary water saturation (modern or in the past). The formation of large crystals is observed in soils with a groundwater level of less than 1 m. This gypsum morphotype is the most common among all the others;

(2) Powdery (with powdery (gypsum meal, gypsum masses) morphotype of gypsum pedofeatures. It diagnoses the occurrence of groundwater close to the surface (<1.1 m) with alternation of the regimes of capillary soil moistening and drying. Coarser-grained ("cracker-type") crystals are formed upon somewhat deeper groundwater occurrence and drier soil, and fine-grained ("marly type") crystals are formed upon shallower groundwater and wetter soil;

(3) Concretionary morphotype is formed in soils with relatively deep groundwater level (~2 m) and periodic soil moistening with flood water. The concretions are represented by gypsum druses of varying degrees of carbonation and are characterized by a complex microfabric.

The formation of large euhedral crystals of gypsum and soluble salts is associated with stable conditions during the slow evaporation of saturated solutions; small crystals are formed upon contrasting moistening regime (wetting/drying conditions) related to fluctuations in the groundwater level. The presence of corrosion on the surface of gypsum crystals and soluble salts indicates the processes of soil desalinization and the predominance of descending water flows; the ideal shape of the crystals indicates the ascending migration of saline water through the capillaries [41]. The formation of a salt crust on the soil surface is due to the evaporative type of soil water regime [44]. It was shown in [37, 42] that the formation of this horizon is possible even when the groundwater salinity is <1 g/L.

A change in the level of saline groundwater leads to a change in distribution pattern of soluble salts. Thus, when the groundwater level decreases, the maximum salt accumulation can shift down the profile, and the upper part of the soil can be washed off from soluble salts. On the example of soils of a subtropical semidesert (Dzhizak steppe), it was shown that a decrease in the level of saline (7–8 g/L) groundwater from 105 to 200 cm leads to desalinization of the upper 18–20 cm and salinization of the second meter of soil within the previously water-saturated horizon in about 20 years [59].

Yamnova [59] gives data on the proportion of open and closed pores in various salt horizons with carbonate, gypsum, and soluble salts. This information makes it possible to evaluate the features of water filtration in the listed horizons. It is shown that in gypsic

and carbonate horizons with relatively low total porosity, closed pores predominate; on the contrary, horizons with the accumulation of soluble salts are characterized by relatively high porosity and the predominance of open pores. Consequently, the carbonate-accumulative and gypsic horizons are characterized by weak filtration properties in comparison with the horizons of accumulation of soluble salts. These data can be taken into account when parameterizing hydrological models for regional conditions. The difficulty of using salt pedofeatures in soils to verify the results of hydrological modeling lies in their often relict nature and the possible lack of connection with modern processes of water transport.

Classification position of soils. In the factor-genetic classification of soils of the Soviet Union [34], special attention was paid to the relationship of soils with the position of the groundwater level. According to this classification, chernozems are formed under automorphic conditions of soil formation with the groundwater depth of more than 6 m; the groundwater level in meadow chernozemic soils is at a depth of 3–6 m; in chernozemic meadow soils, at 1.5–3 m; and bog soils are formed at the groundwater level of less than 1.5 m. This made it possible to use information from soil maps (interpretation of the groundwater level in soils according to their classification position) to parameterize the equation of continuity of the stationary state in the kinematic wave function or soils of the forest-steppe of the Oka–Don Lowland [119].

The study of the water regime of soils at research stations made it possible to characterize the dynamics of soil moisture, the level of groundwater occurrence, the characteristics of water recharge, desiccation, and other properties of various types of soils [4, 6, 36, 43, 55]. For example, it is shown that the water regime of ordinary and southern chernozems belongs to the nonpercolative type (there is no through wetting) for the observation period from 1950 to 1960s. The work [5] shows a change in the water regime of these soils towards greater humidization in the period from the 1970s to the present, when through wetting of the soil profile has been observed periodically. It should be noted that the change in the water regime of soils during the specified period has not led to significant changes in the classification-significant soil properties, and these soils are still diagnosed as ordinary and southern chernozems. Hence, the unambiguous diagnosis of the water regime of soils according to their classification position is debatable.

In the American *Keys to Soil Taxonomy* [82], a large number of suborders and even great groups of soils are distinguished on the basis of their moisture regime characteristics. In this case, the moisture regime is determined for some part of the soil profile, called the moisture control section (MCS); its upper limit is determined by the depth of soil wetting by 2.5 cm of rainfall during 24 h, and the precipitation

over a 24-hour period, and the lower limit is determined by the depth of soil wetting by 7.5 cm of rainfall during 48 h. On average, for loamy soils, MCS is confined to depths of 10–30 cm; for sandy soils, it is 30–90 cm. Aquic, Aridic (Torric), Udic, Perudic, Ustic, and Xeric soil moisture regime classes are separated. The latter five classes are determined by the number of days with a dry or wet MCS. This information can be used to verify the results of hydrological modeling of soil water regime features.

LABORATORY METHODS FOR DIAGNOSTICS OF THE PARAMETERS OF SOIL WATER REGIMES

Indicators of soil moisture and air porosity. Soil moisture can be expressed as absolute or relative water contents, or in terms of soil hydrological constants. According to Rode [44], soil waterlogging takes place when the water content is above the field water capacity; when this period is long, this soil can be attributed to hydromorphic soils. Romanova [45], based on many years of research and generalization of data from the network of weather stations suggested that a number of days with the soil water content in the upper 20 cm above the field water capacity might be used as a criterion of hydromorphism for Belorussian soils [45]. This criterion was applied to subdivide soils into the groups of automorphic, semihydromorphic, and hydromorphic soils.

The air porosity index is the reciprocal of the volumetric water content. On the basis of this indicator (calculated as the difference between the total porosity and the actual volumetric water content), Shein [58] and Zaidel'man [26] identified the threshold values of ecological waterlogging, that is, such a state of the water and air regime of soils, upon which a negative impact on crop yields is observed. According to Shein [58], ecological waterlogging occurs at an air porosity of 10% or less, according to Zaidel'man [26], at 8% in the plow horizon and 6% in the subsoil.

Antsiferova [2] showed that burozems with different degrees of gleying significantly differ from one another in the duration of the period with air porosity of <10% in the upper soil meter regardless of the conditions of atmospheric moistening (low, medium, or high) for the periods considered in the work. At the same time, the duration of the period with soil water content in the layer of 0–100 cm above the field water capacity differed significantly for burozems with different degrees of gleying only under conditions of medium and increased moistening; in dry years, the water content below the capillary rupture moisture in the same layer was more indicative.

Thus, indicators based on measuring soil water content or air porosity can characterize both the current state of soils (for example, the presence of waterlogging or drying at the moment) and the characteris-

tics of the water regime of soils in general. Data on water content of air bearing porosity can be used both for parametrization of hydrological models and for verification of the obtained results.

Forms and content of iron in soils. The Schwertmann (K_{sch}), Vodyanitskii (K_v), and Bodegom (K_b) criteria are based on different ratios of iron contents extracted from soils with the Tamm (Fe_{ox}) and Mehra–Jackson (Fe_{dit}) reagents and have the form [9]:

$$K_{sch} = \frac{Fe_{ox}}{Fe_{dit}},$$

$$K_b = 0.19Fe_{ox} - 0.028(Fe_{dit} - Fe_{ox}),$$

$$K_v = \frac{Fe_{ox}}{Fe_{ox} + Fe_{dit}}.$$

The fundamental difference between these criteria is that in the case of K_{sch} and K_b , the results of parallel extraction from soils are used, whereas K_v implies the sequential extraction of oxalate-soluble and dithionite-soluble iron. The higher the value of the criteria, the higher the degree of soil waterlogging. The question remains why the proposed ratios of extracts by Tamm's and Mehra–Jackson's reagents are sensitive to the degree of soil waterlogging. As explanations, assumptions have been made about an increase in the proportion of amorphous compounds extracted by the Tamm reagent in gleyed soils [19, 62] and high sensitivity of the Tamm reagent to the presence of Fe^{2+} catalytically accelerating the dissolution of Fe^{3+} -(hydr)oxides [10, 12].

Schwertmann's criterion is used to diagnose the processes of waterlogging of soils that form in a wide range of physiographic conditions [1, 9, 53]. There is no unified scale for the values of the relationship between the criterion and soil hydromorphism [9]; however, for each region, its own semiquantitative scales should be created. It has been shown that this criterion makes it possible to identify soils with surface moistening rather than with ground moistening more reliably [31] and characterizes the actual processes of soil waterlogging [1].

Like the Schwertmann criterion, the Bodegom criterion is used for soils formed in various landscape conditions [1, 9, 114]. This criterion makes it possible to quantify the content of Fe^{2+} formed as a result of biogeochemical processes [114]. In addition to the features of hydromorphism, the Bodegom criterion can characterize the intensity of methane release from soils [73].

The criterion of Vodyanitskii [9] has not yet become widespread in the diagnosis of soil hydromorphism; at the same time, it is very promising due to the presence of a single universal scale that relates the degree of manifestation of hydromorphism and the values of the coefficients, as well as the use of parallel extractions increases the accuracy of determining dif-

ferent forms of iron compounds. Waterlogging is absent in soils at K_v values < 0.15 ; values > 0.6 correspond to waterlogged soils. Like the criteria listed above, the Vodyanitskii criterion can be used for soils formed under different environmental conditions.

Zaidel'man [27] proposed a coefficient equal to the ratio of the iron content in the pyrophosphate extract (according to Bascomb, organomineral iron) to the iron content in Tamm's extract; the higher the value of the coefficient, the longer the period of waterlogging. This coefficient has been successfully used for soils formed on Permian red-earth deposits, the color features of which mask the morphochromatic signs of soil waterlogging [27].

Another diagnostic criterion for waterlogging processes is the index of iron oxidation (K_o), determined, in particular, from the results of Mössbauer NMR spectroscopy according to the formula [9]:

$$K_o = \frac{Fe^{3+}}{(Fe^{3+} + Fe^{2+})}.$$

The higher the value of the index, the less pronounced are the processes of gleying. This indicator can be used in both humid and semiarid climates [9]. The listed indicators characterize the features of the actual water regime of soils and can be used to verify the modeled parameters of hydrological processes corresponding to the current conditions of soil formation. Features of the profile/landscape distribution of indicators make it possible to rank the studied objects according to the duration of the waterlogging period.

Chemical composition of soil pedofeatures. According to [29, 78], the ratio of iron to manganese in nodules (determined in 1 n sulfuric acid extract) depends on the duration and intensity of the period of soil waterlogging. This coefficient (the ratio of iron to manganese in nodules composition) is called the "swampiness" coefficient [29] or the Zaidel'man coefficient [9]. The values of this coefficient and the corresponding levels of waterlogging were obtained for various loamy and clayey deposits (cover, moraine, two-layered, lacustrine-glacial, Permian, and some others) and published in [22–24, 32]. The absolute values of the coefficient can vary over a wide range (from 0.001 to 5000); the higher the value of the coefficient in the considered series of soils formed under similar lithological and climatic conditions, the higher the intensity and duration of soil waterlogging. It was shown in [46] that this coefficient can be applied not only to nodules but also to other ferromanganese formations, except for rhizoconcretions. Note that the waterlogging coefficient is not applicable for soils on calcareous parent rocks.

The content of mobile forms of iron and manganese, as well as the fractional composition of nodular phosphates, make it possible to diagnose the source of soil overmoistening—surface or ground, as was

shown in the case of the soils of the north of the Tambov Plain [26]. In addition, Zaidel'man et al. [24] noted that the ratio of iron extracted by the Mehra–Jackson reagent from the clay fraction of coatings (cc) and clay fraction of parent material (cpm) correlates with the intensity and duration of soil waterlogging processes. The following ratio has been proposed:

$$\frac{(Fe_{cpm} - Fe_{cc}) \times 100}{Fe_{cpm}}.$$

Determining this coefficient is quite laborious, both in terms of sampling (mainly coatings) and chemical-analytical work (fractionation of the samples according to particle-size distribution is necessary), so this coefficient is rarely used.

Thus, the considered coefficients can be used to verify the duration of the period of soil waterlogging and to identify areas with different moisture conditions. These criteria are suitable only for soils with a contrasting water regime with the presence of periods of desiccation and waterlogging; they can characterize not only modern, but also relict moistening conditions, which makes it debatable their use as criteria for verifying the results of hydrological modeling.

Mineralogical composition of soils. The presence of hematite minerals is associated with the predominance of a pronounced oxidative environment (absence of waterlogging); ferrihydrite, ferroxhyte, lepidocrocite, and goethite, on the contrary, attest to pronounced waterlogging [8, 66, 101]. Green rust is a specific iron mineral that forms only in waterlogged sandy soils or peat. This mineral is very unstable and sensitive to redox conditions: it is present in soils only during reduction periods and disappears during oxidation periods [16]. Iron hydroxides can be formed both in the iron–manganese nodules and in the soil plasma forming accumulations or being disseminated in the soil mass. To determine the mineralogical composition, Mössbauer NMR spectroscopy, X-Ray diffraction, and transmission electron microscopy coupled with electron microdiffraction can be used.

Ferrihydrite is found in both taiga and steppe soils. High values of the redox potential and an abundance of organic ligands are favorable for the formation of ferrihydrite. Bacteria are involved in the formation of ferroxhyte, (like ferrihydrite) in soils; ferroxhyte formation is associated with variable Eh values under conditions of organic matter deficiency. Ferroxhyte is found only in acidic soils. Its predominance in the composition of soil nodules indicates actual (modern) processes of soil waterlogging. Like ferroxhyte, lepidocrocite is formed under a variable redox potential and a lack of organic carbon, but mainly in cold climates and abiotically. It was shown in [102] that this mineral is mainly confined to iron nodules with a high content of clay minerals.

Goethite is the most abundant iron hydroxide mineral in soils; its formation occurs abiotically, and the

high content of organic carbon contributes to the formation of goethite in the nodules. At the same time, a decrease in the proportion of aluminum in the composition of goethite is a response to soil waterlogging. The predominance of goethite in the composition of concretions (compared to other iron minerals) indicates the inherited gley, the formation of these pedo-features in earlier periods of soil formation [9].

A quantitative indicator based on the mineralogical composition can be the criterion of iron hydroxide genesis (K_g) determined according to the formula [9]:

$$K_g = \frac{\text{iron hydroxides}}{(\text{iron hydroxides} + \text{iron oxides})}.$$

The content of iron hydroxides is interpreted as the content of goethite, and iron oxides correspond to hematite.

In our opinion, the greatest interest in using data on the mineralogical composition of soils for the purpose of verifying the results of hydrological modeling lies in the identification/confirmation of genetic hypothesis about the formation of secondary minerals in soils.

Characteristics of soil organic matter. Soil moisture affects the activity and abundance of microbial biomass [103] and contributes to the destabilization of organic matter making it more accessible to soil microorganisms [81]. Therefore, soil moisture is one of the leading environmental parameters controlling the mineralization of soil organic matter and newly arriving plant residues [88, 107]. Thus, the characteristics of the process of mineralization of organic matter or the features of its composition can act as a diagnostic indicator of changes in the parameters of the water regime of soils. It should be noted that the effect of soil moisture on the rate of mineralization is not linear and has a dome-like character: the minimum values occur both under conditions of the lack and a clear excess of moisture [103].

The emission of carbon dioxide from the soil is one of the important indicator of the rate of mineralization of organic matter, which is sensitive to changes in soil moisture in the short term [106]. To characterize the long-term effect of the lack of moisture/normal moistening or overmoistening of soils on the mineralization of organic matter, a range of indicators has been proposed in [51, 52, 79]: the content of potentially mineralizable carbon; easily, moderately, and difficultly mineralizable fractions of active organic matter. It has been shown that with an increase in the water content of chernozems, chestnut, and gray forest soils from 10 to 40%, the degree of mineralization of soil organic matter increases [52]. At the same time, the fraction of readily mineralizable organic matter is the most sensitive to changes in moisture conditions; its share in the structure of active organic matter increases by a factor of 1.5 with an increase in the soil moisture [51].

The dynamics of the content of the water-soluble fraction of soil organic matter can also act as an indicator of the degree of soil moistening. Using a model experiment as an example, the authors of [65] showed an increase in the content of the water-soluble fraction of organic carbon extracted from the forest litter when it is moistened. Temporary surface seasonal waterlogging contributes to the fixation of the water-soluble fraction of organic matter in the soil [33].

In soils of the tundra and taiga zones, excessive moisture significantly affects the accumulation of humus compounds and affects their molecular structure [40]. Constant waterlogging can cause the process of peat accumulation; waterlogging during a larger part of the growing season results in the formation of mucky horizons [18, 91].

Zaidel'man [27] demonstrated a relationship between the optical density of the pyrophosphate extract extracted from the humus horizon and the duration of waterlogging for calcareous soils with weak visual signs of hydromorphism (due to the specific features of parent material). Stepantsova and Krasin [49] proposed a coefficient for the degree of soil hydromorphism based on the ratio of the optical densities of alkaline and pyrophosphate extracts from the material of the arable soil horizon. This coefficient has been successfully used to estimate the duration of surface waterlogging of soils [31, 39], to diagnose soil waterlogging, and the level of groundwater [49] in forest-steppe soils. The coefficient of the degree of hydromorphism reflects the quality of soil organic matter, since it is the ratio of fractions I (free and bound with sesquioxides) and II (bound with calcium) of humic substances. Probably, other indicators of the quality of organic matter, such as the ratio of humic acids to fulvic acids, humification index, optical density of humic acids, etc. [47] can also be used as diagnostic indicators of soil water regime.

Thus, the features of soil organic matter can be used to verify the results of hydrological modeling for both short (hours) and long (months–years) periods.

Isotopic composition of soil water. Stable isotopes of soil water—oxygen ($^{16}\text{O}/^{18}\text{O}$) and hydrogen ($^1\text{H}/^2\text{H}$; ^3H)—are natural tracers for studying the features of water movement in soils. The inclusion of this section in the review is due to the high prospects of using data on the isotopic composition of pore and lysimetric waters to verify the parameters of the water regime of soils and the poor coverage of this issue in Russian literature.

Water evaporation in the soil body and from the soil surface leads to a change in the ratios of stable isotopes: since the molecular weight of $^1\text{H}_2^{18}\text{O}$ (20.015 g/mol, [76]) is greater than that of $^1\text{H}_2^{16}\text{O}$ (19.017 g/mol [76]), a slightly more intense evaporation of lighter molecules takes place. Thus, the isotopic composition of soil water becomes different from the isotopic composition of precipitation in the direc-

tion of its enrichment in heavy isotopes. This process is called evaporative fractionation. In [71], which was devoted to a meta-analysis of the isotopic composition of water in pore space, it was found that evaporative fractionation processes are typical up to a depth of 3 m in an arid climate, 0.5 m in a Mediterranean climate, and up to 0.2–0.3 m in soils of a temperate climate.

In general, plants do not affect the isotopic composition of water during root absorption of moisture, with the exception of some species of halophytes [86] and xerophytic shrubs [69]. In this case, the evaporation of water through the stomata of plants leads to the fractionation of the isotopes [118]. Based on special physical and mathematical models, it is possible to determine the ratio of water of transpiration genesis to water evaporated from the soil surface [95, 117, 121]. In addition, the analysis of the isotopic composition of water can be used to identify the zone of influence of water uptake by roots [97].

In [108], a review of studies into the features of hydrological processes in soils by the methods of isotopic analysis of stable isotopes of water is given. In particular, the possibility of determining the sources of moisture inflow into the soil (with atmospheric precipitation or groundwater) and their ratio [83, 93], identifying the preferential radial and lateral soil flows [98], the duration of water staying in soils, and the rate of water infiltration [109, 111].

CONCLUSIONS

The processes of soil water redistribution in radial and lateral directions contribute to the formation of a number of specific properties in soils with different formation times and resistance to changes in the environmental factors. The use of soil properties in order to verify the results of hydrological modeling of soil water regime parameters is based on a comparison of the profile distribution patterns of indicators and modeled parameters. A similar principle can be applied to verify the features of lateral soil flows—properties of contiguous soil areas can be used as indicators of the process. The use of soil properties as indicators of the processes of soil water transfer contributes to the deepening of knowledge about the genesis of soils and individual soil properties.

Based on a standard field description of soils, it is possible to identify zones in the soil profile (their depth and thickness) characterized by constant (by color) or periodic (by color patterns, the presence of pedofeatures with polyvalent metals) water stagnation. In landscapes with balanced or insufficient moistening, the position of the soil effervescence line and the detection of carbonate pedofeatures make it possible to identify the zone of the highest moisture turnover (the presence of both ascending and descending moisture flows). Pedofeatures associated with gypsum and soluble salts allow us to make assumptions about the

depth of the capillary fringe level. Clay coatings, root paths, and animal burrows are the markers of preferential flows. Marly horizons (calic) and iron-plated (rudzak, iron-ore) horizons may be the result of lateral soil water flows. The morphology and size of carbonate, gypsum, and salt pedofeatures correspond to the conditions of their crystallization (relatively slow or intense), and also allow us to make assumptions about the saturation of solutions. Salt pedofeatures may be inherited by the soil from previous stages of soil formation; determination of the modern or relict genesis of pedofeatures by field methods is difficult, and additional studies are needed (for example, micromorphological), which complicates the use of these criteria in order to verify the results of hydrological modeling and makes them debatable.

Analytical indicators for diagnosing the parameters of the water regime are aimed at assessing the quantitative ratios of hydrological processes in the soil; Basically, they make it possible to characterize the duration and intensity of the period of waterlogging. These indicators include coefficients based on the ratios of various forms of metals with variable valence (Schwertmann, Vodyanitskii, Bodegom, and Zaidelman's criteria) and the composition of soil organic matter. The content of mobile forms of iron and manganese and the fractional composition of phosphates in iron nodules, as well as the ratio of optical densities of alkaline to pyrophosphate extracts, make it possible to diagnose the source of soil overmoistening: surface or ground. The use of the mineralogical composition of soils and the micromorphological structure of salt pedofeatures in order to diagnose the water regime of soils can make it possible to distinguish between modern and relict features of waterlogging.

Analytical methods for diagnosing the parameters of the water regime are characterized by varying degrees of labor intensity, including both the complexity of sampling (e.g., sampling of coatings from ped faces), sample preparation, and the analytical procedure per se. The values of most diagnostic indicators depend, among other things, on the lithological features of soil-forming materials, which necessitates the creation of local and regional scales of indicator values to assess the degree of soil waterlogging. For example, the same values of the coefficient of the degree of hydromorphism can correspond to soils with different moisture conditions. The Schwertmann criterion is the most commonly used analytical indicator for diagnosing waterlogged soils, the possibility and effectiveness of which has been shown for a wide range of soils.

The review shows that verification of the results of hydrological modeling on the basis of data on the soil properties is potentially possible for the processes of downward and lateral water flows, physical evaporation of moisture, transpiration, evapotranspiration, and capillary rise. Soil properties determined both in the field and in the laboratory make it possible to verify

the results of modeling the duration of soil waterlogging for a wide temporal range.

The arrangement of the indicators considered in this review from the point of view of their genetic interpretation, the characteristic time of formation of the equilibrium state, and their resistance to changes in the environmental parameters, and their use for verification of the results of simulation of hydrological processes cannot be universal for all geographic settings and must be adapted to the zonal and provincial features of the soil cover.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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